



WAREHOUSE LAYOUT OPTIMIZATION

A Simulation Study of Storage Assignment Policies at the Schneider Electric Manisa Plant Warehouse

Deniz Etensel · Deniz Ege Memetoğlu · Nehir Konyar · Elif Bostancı · Yiğit Kaçar · Sümeyra Pulça | Supervisor: Dr. Oktay Karabağ

44%
lower kitting lead time
12.37 → 6.91 min

1.3%
reach-truck utilization
down from 21.7%

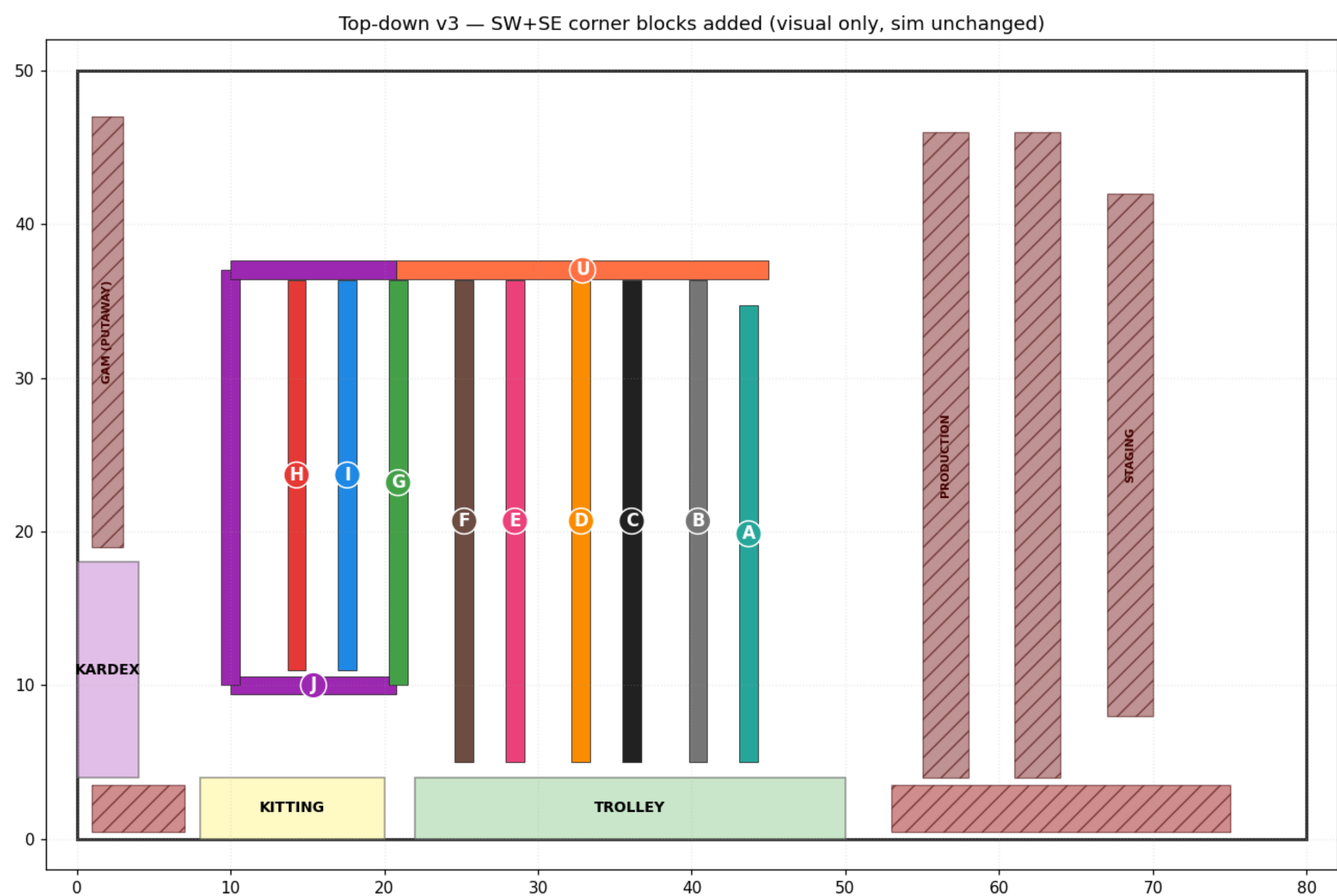
3.10
Cohen's d effect size
very large ($\gg 0.8$)

0
physical changes needed
re-slotting only

1. The Problem

The Schneider Electric Manisa plant feeds several production lines through a **kitting** warehouse: every order is a complete set of components assembled before production. The current SAP ABC/FMR tool ranks materials by *aggregate* demand and ignores **which line** a material serves, so the components of one kit scatter across different corridors and levels. When part of a kit sits high on the racks, the operator must wait for one of the **7 shared reach trucks**; these requests queue, and the waiting flows straight into kitting lead time.

Can a smarter slotting policy remove this waiting without any physical layout change?



Modelled warehouse: 11 rack rows (A–J, U); kit corridors alternate with reach-truck aisles.

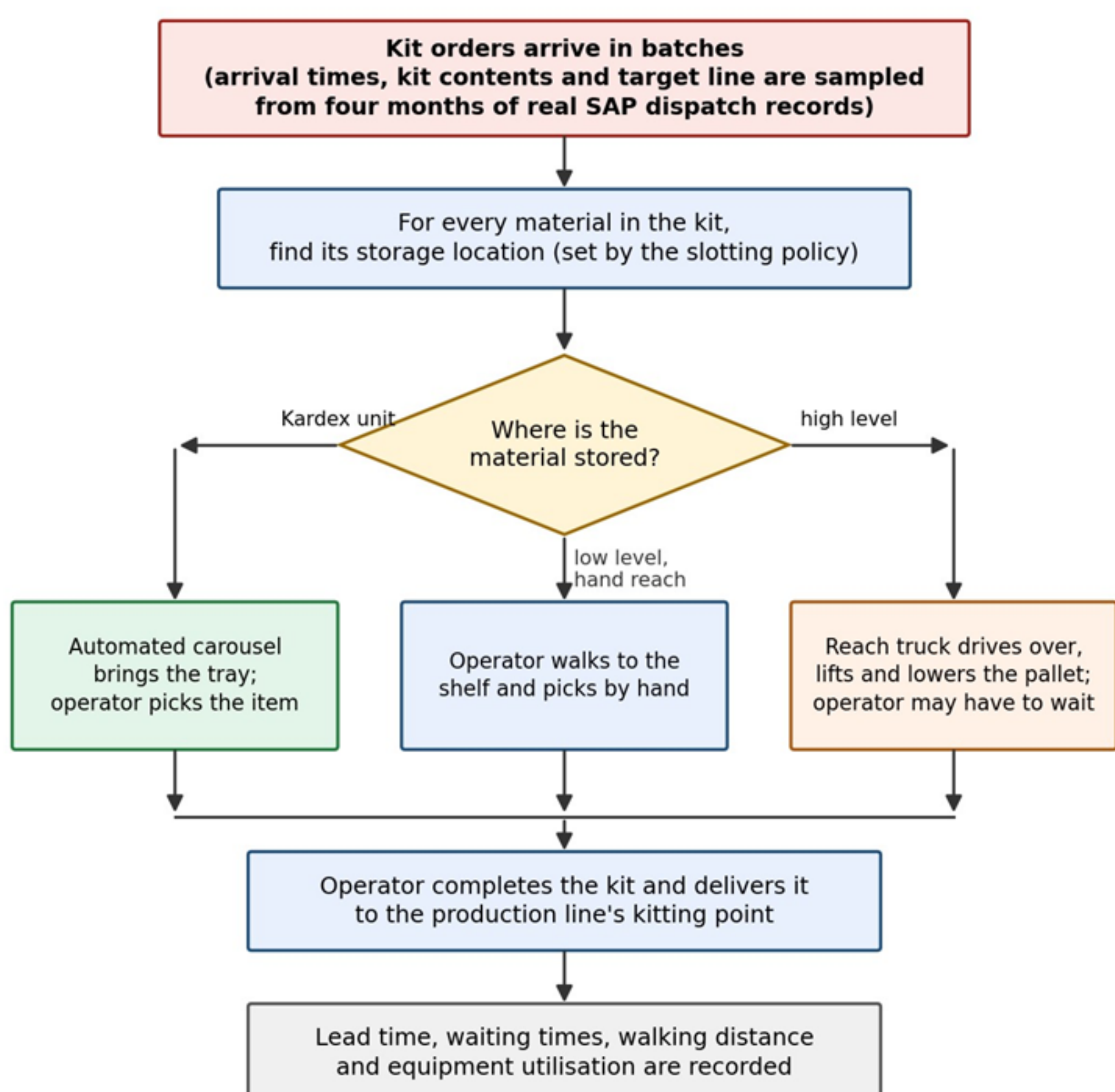
2. Method

A **discrete-event simulation** (Python / SimPy) driven by **four months of real SAP ZWM92 data** (40,804 kit orders, 103 active days) and calibrated with an **F400 video time-motion study** (2,319 micro-events). Arrivals use empirical batch sizes and gaps; service times are lognormal, fitted by $\sigma = \sqrt{\ln(1 + CV^2)}$.

Six storage policies, all under identical demand ($N = 20$, common random numbers):

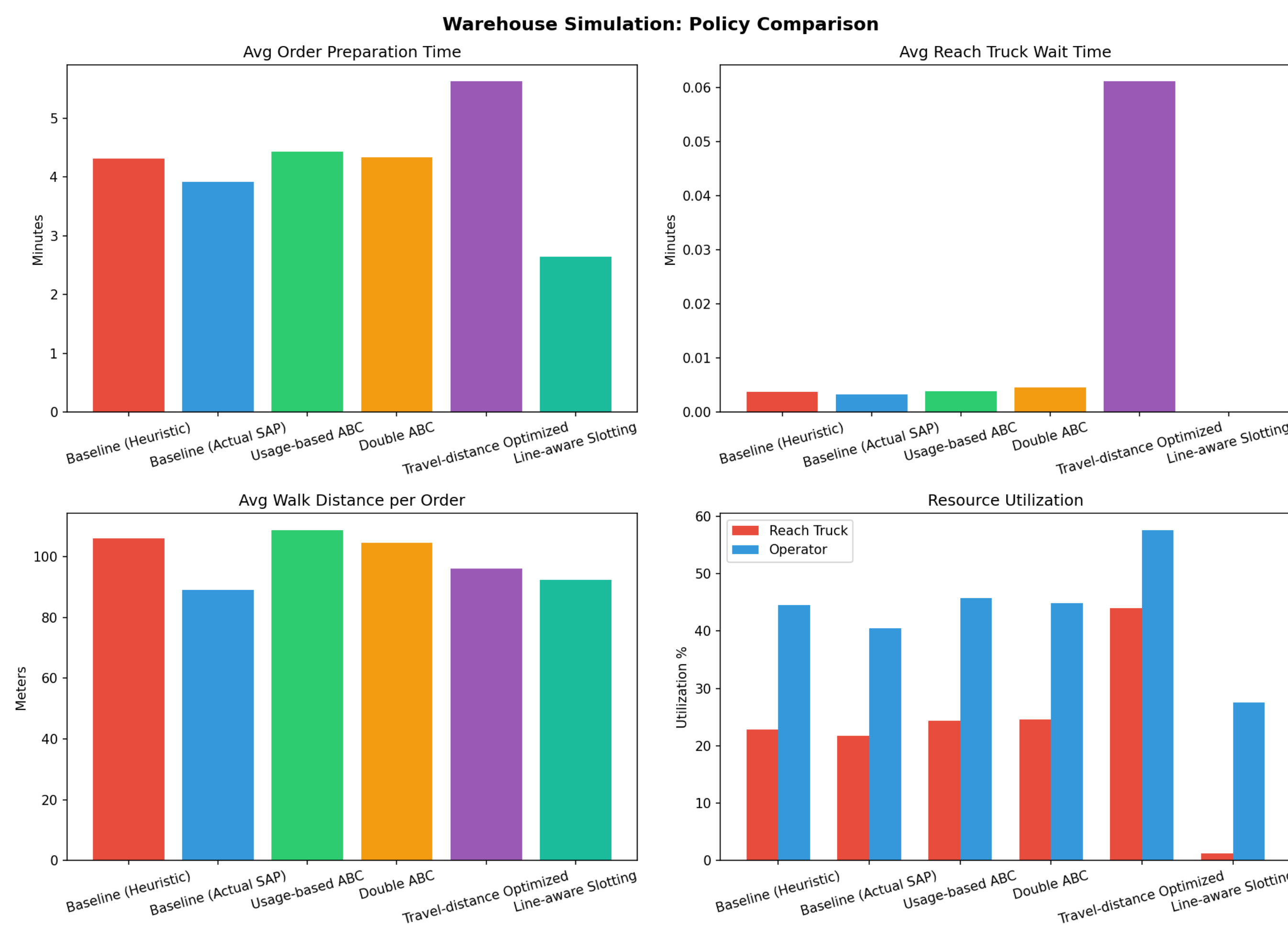
- Baseline – Actual SAP · Capacity heuristic
- Usage-based ABC · Double ABC · Travel-distance Optimized
- **Line-aware Slotting (proposed)**

How the simulation handles one kit order



3. Results

Across $N = 20$ replications, **Line-aware Slotting** cut mean lead time from 12.37 to **6.91 min** ($p < 0.001$) and reach-truck use from 21.7% to **1.3%**. Walking distance barely moved (89.0 → 92.4 m), so the gain comes from **ending reach-truck queueing**, not shorter walks.



Policy comparison across four KPIs.

Policy	Lead (min)	RT util.
Line-aware Slotting	6.91	1.3%
Baseline (Actual SAP)	12.37	21.7%
Baseline (Heuristic)	14.71	22.8%
Double ABC	14.74	24.6%
Usage-based ABC	15.24	24.4%
Travel-distance Optimized	24.60	44.0%

4. Statistical Evidence

One-way **ANOVA** rejects equal means for lead time, waiting, utilization and walk distance ($p < 10^{-8}$); throughput is unaffected ($F = 0.01$). The CRN **paired t -test** gives Line-aware vs. SAP = -5.46 min ($t = -17.47$, $p = 3.7 \times 10^{-13}$), with effect size **Cohen's $d = 3.10$** . Under a **doubled-load stress test** the advantage grows from 5.46 to **20.7 min**, so the policy matters most when the system is busiest.

5. Conclusion

Slotting **line by line** and keeping fast movers at **hand-reachable levels** cuts kitting lead time by about **44%** and nearly removes reach-truck dependence, **with no new equipment and no physical layout change**. In a kitting warehouse with shared reach trucks, the real lever is **resource queueing**, not travel distance.